

ZKFOCIL

GEORGE, THOMAS, BENEDIKT

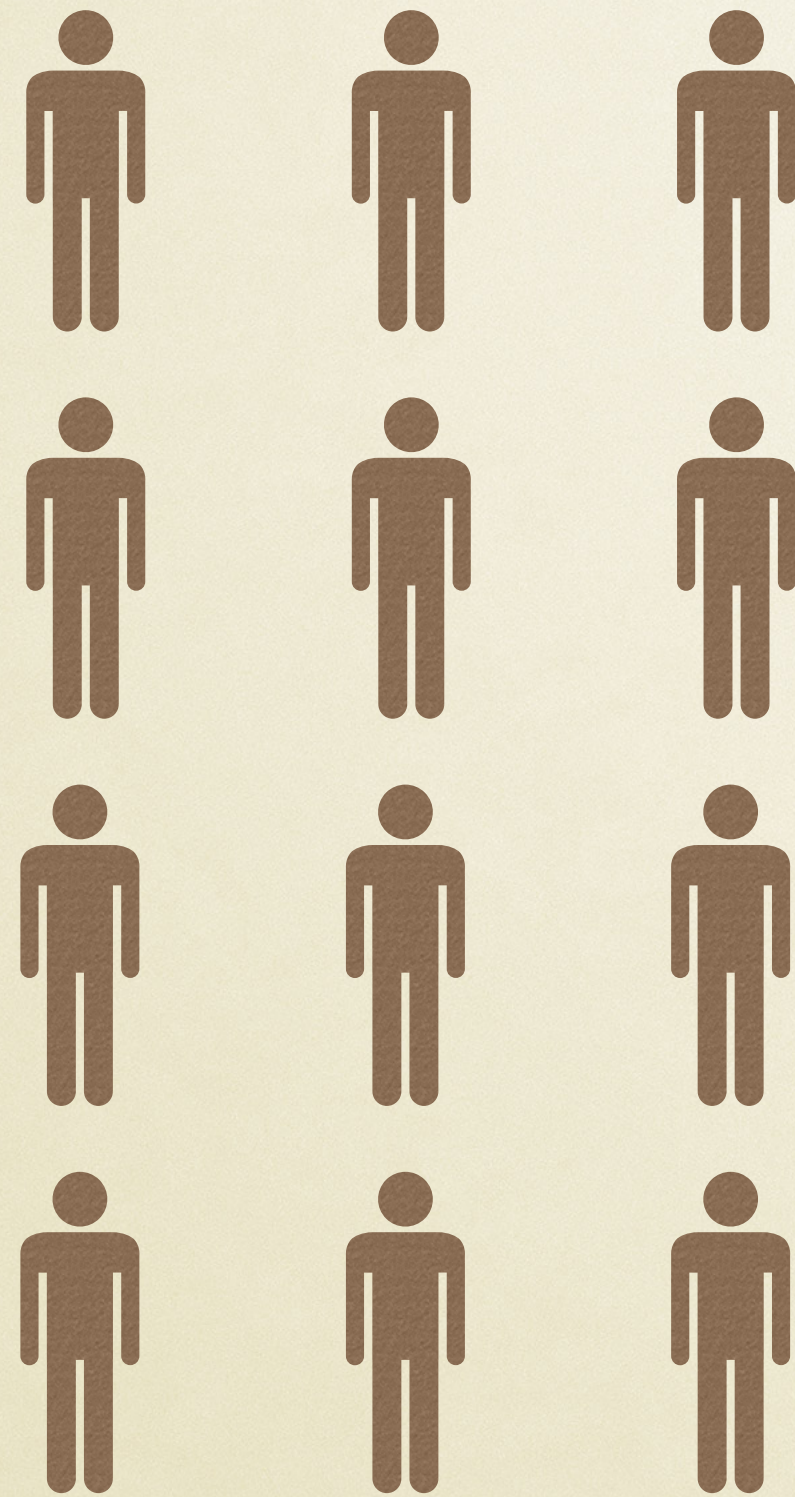


TALK BY BENEDIKT

Sila Foundation Cryptography Research

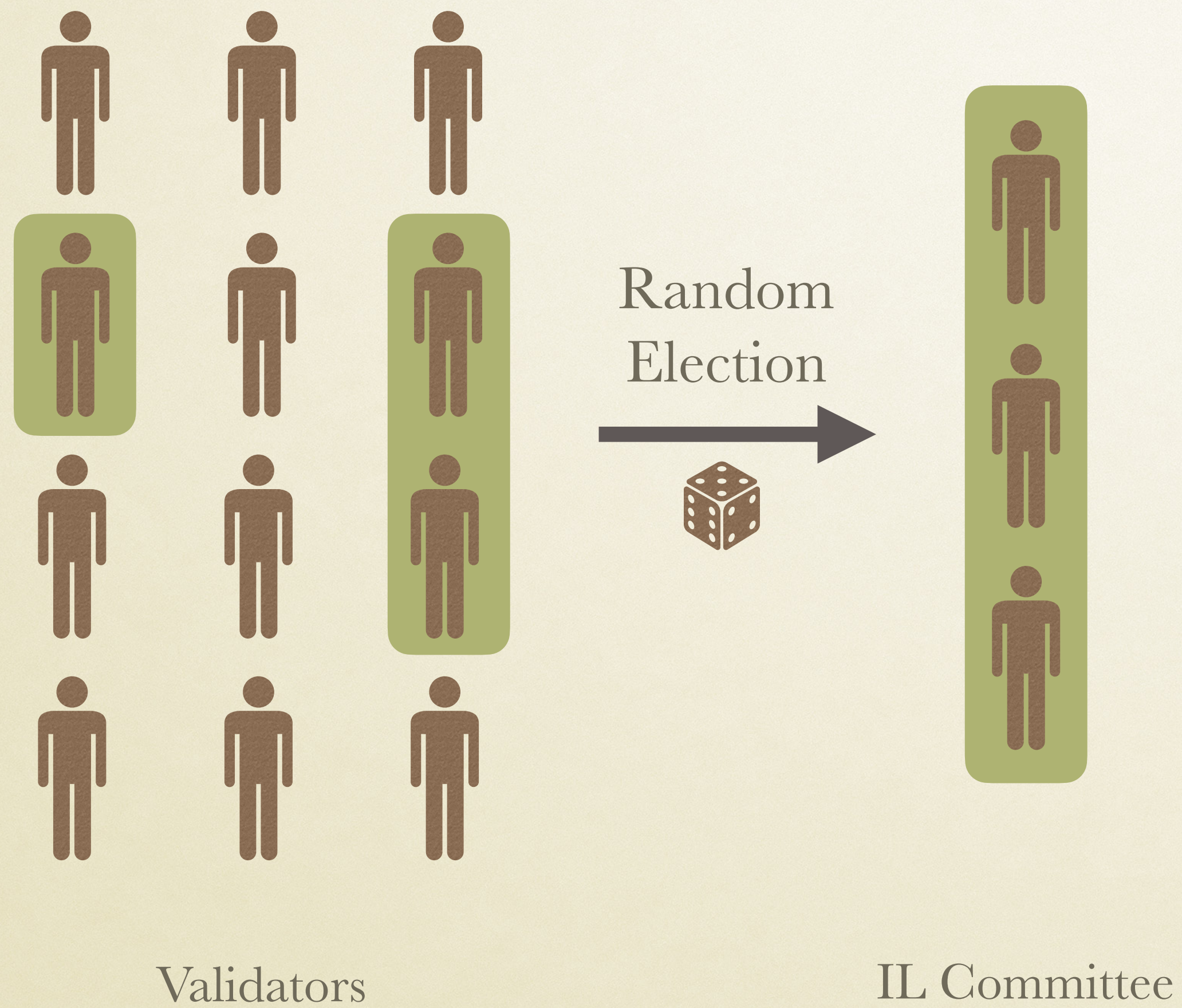
RECAP: FOCIL

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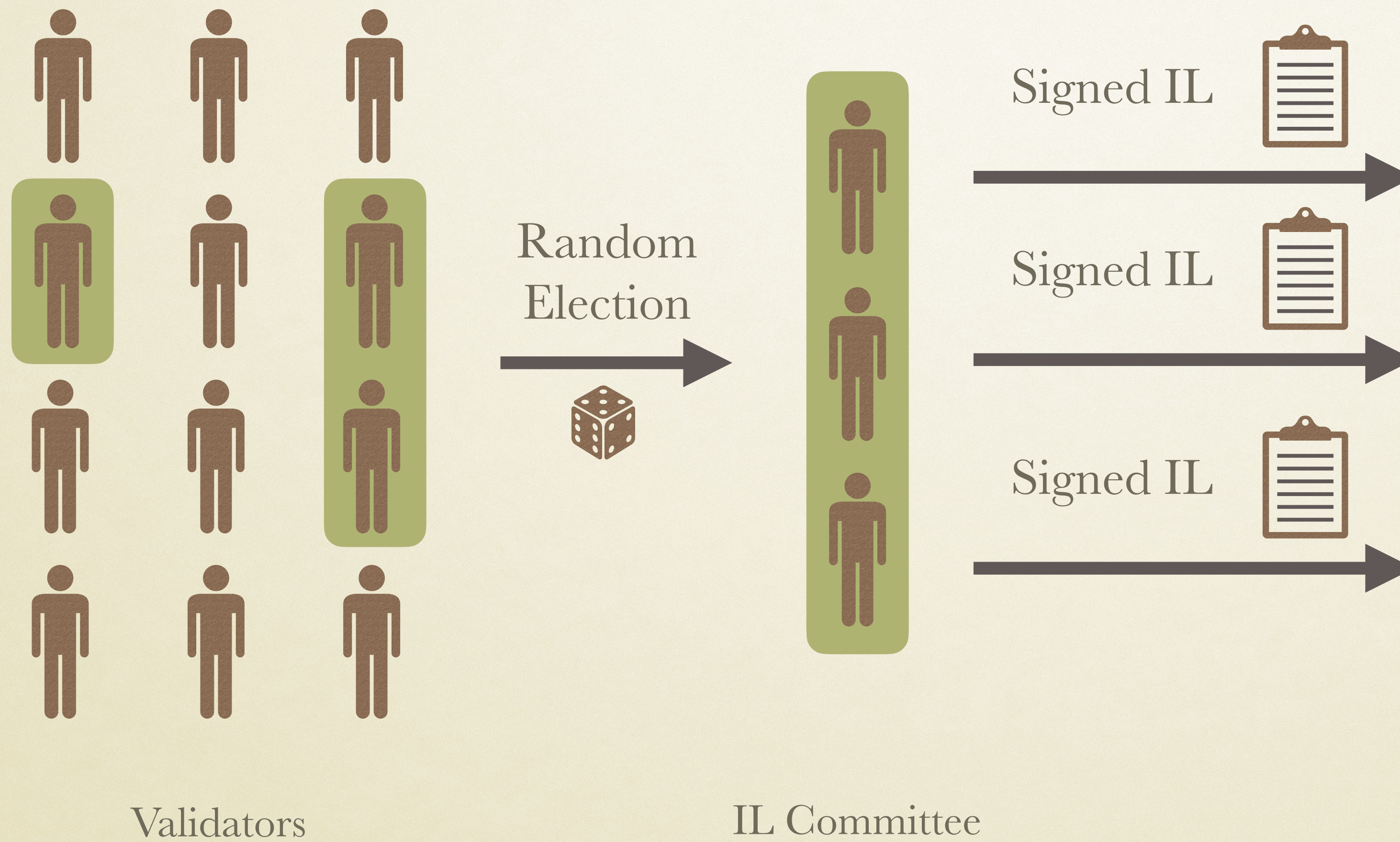


Validators

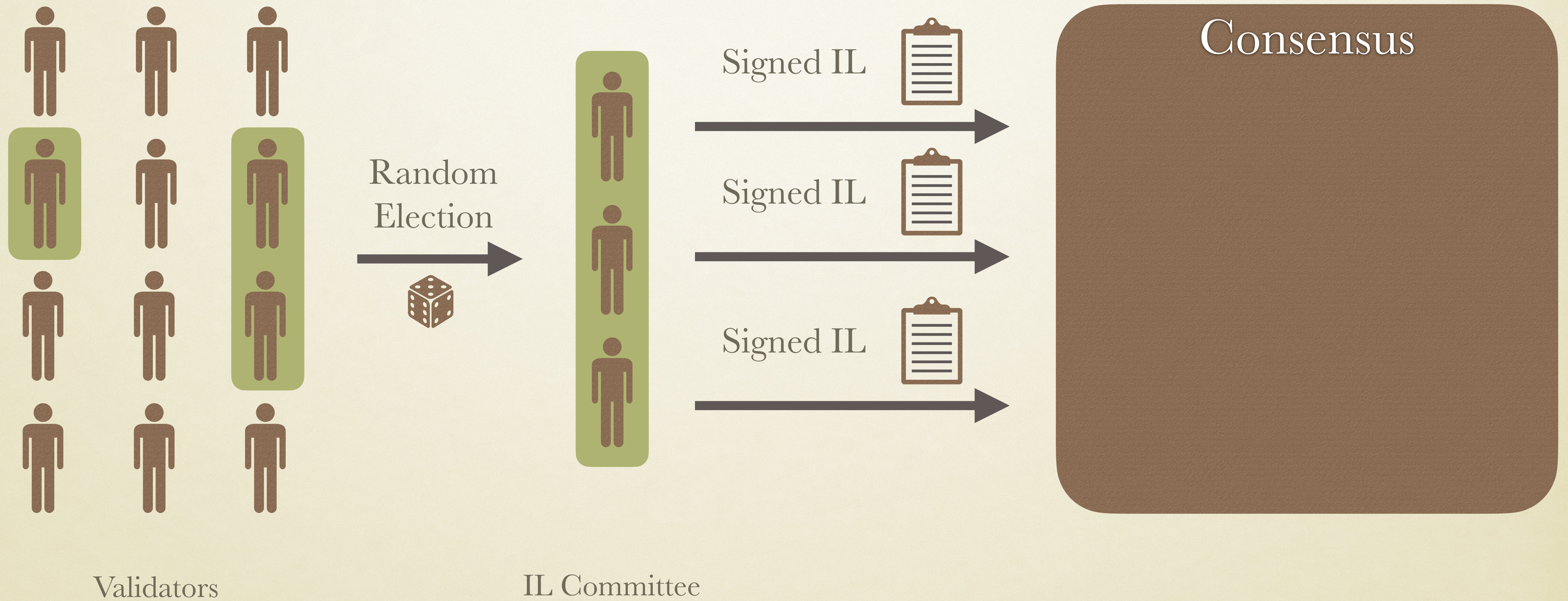
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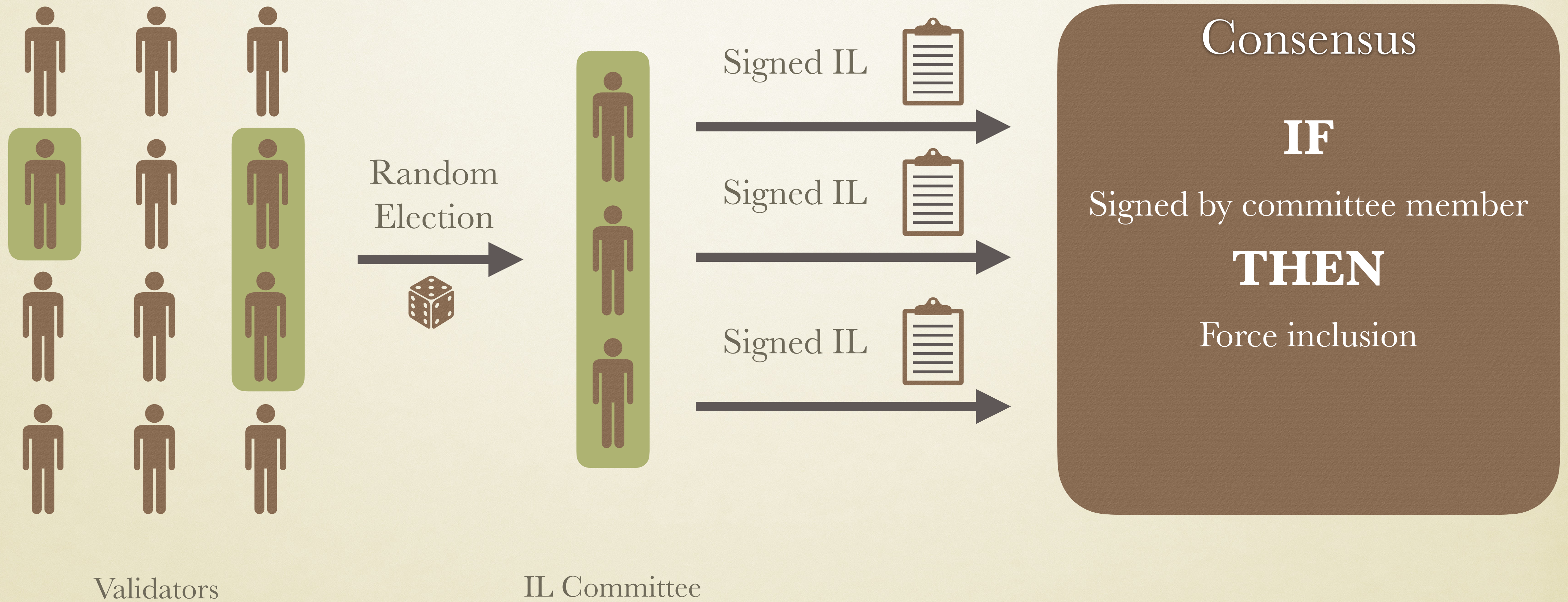
RECAP: FOCIL



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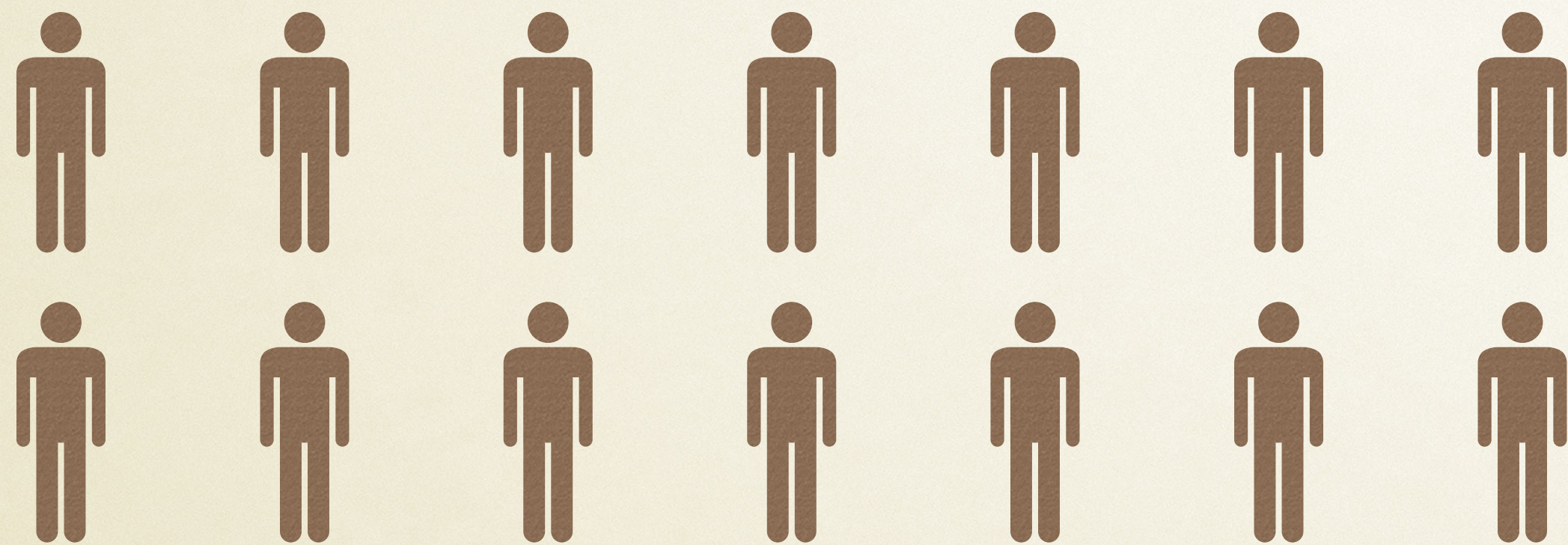
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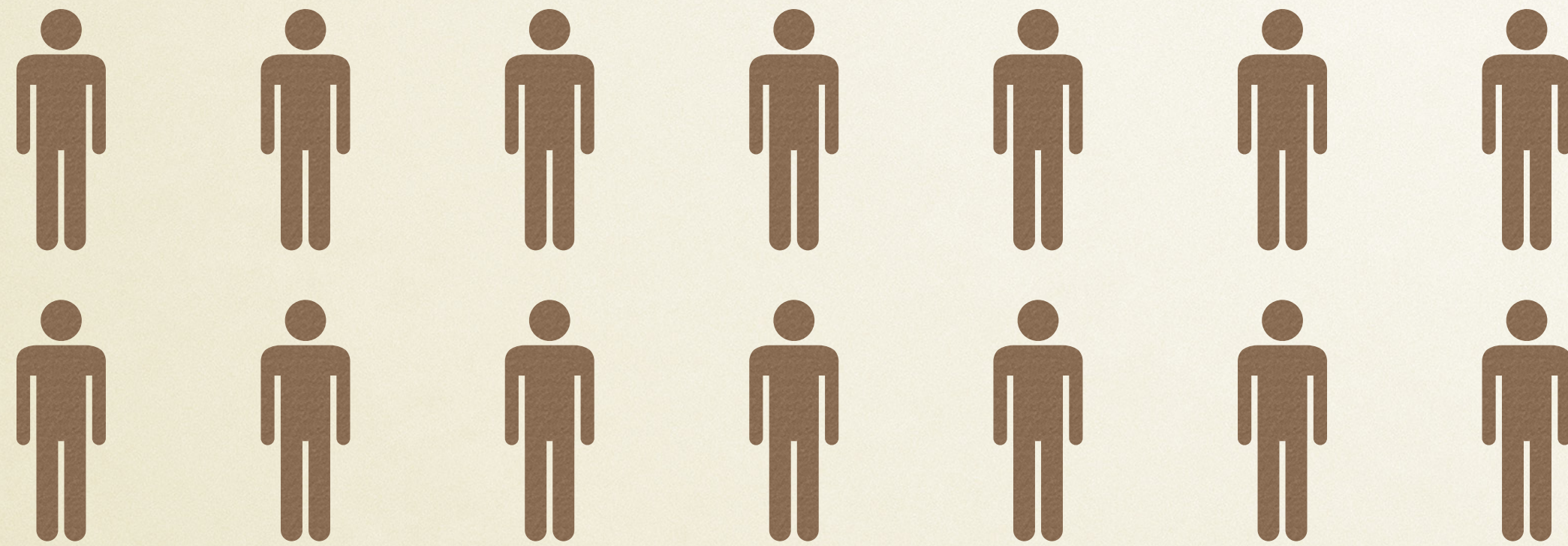
Validators



Inclusion Lists

RECAP: FOCIL

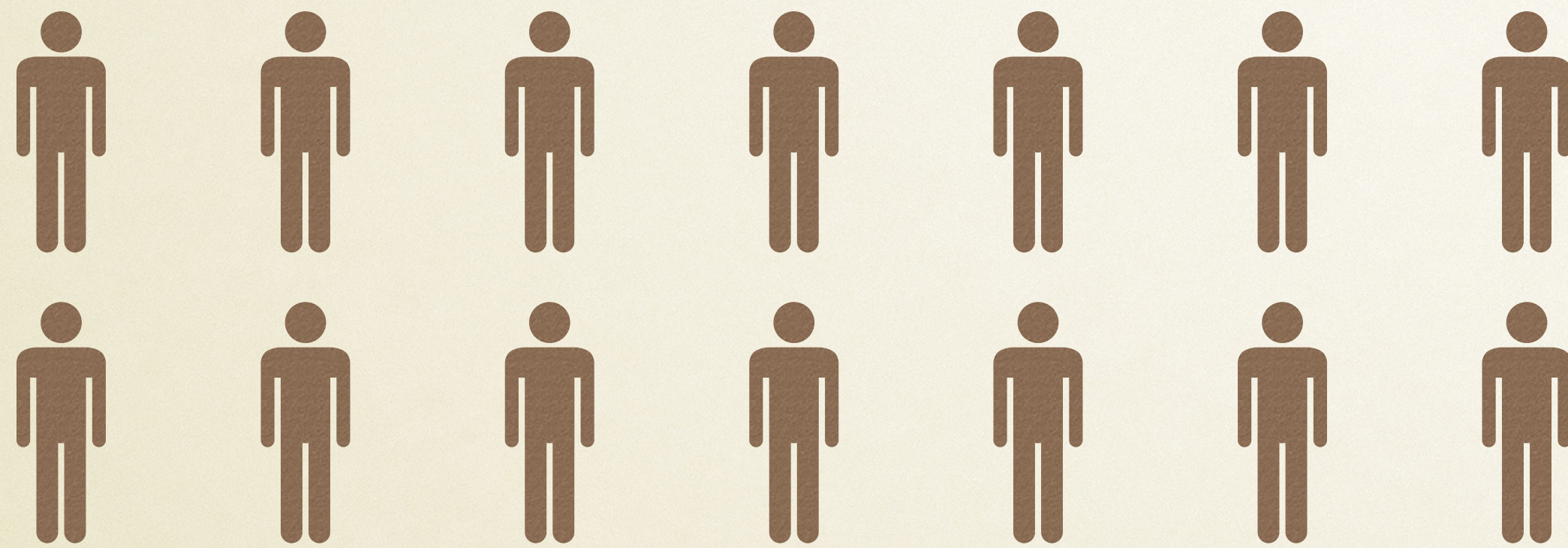
Validators



Inclusion Lists

RECAP: FOCIL

Validators



Inclusion Lists

Can we hide the link?

GOALS OF ZKFOCIL



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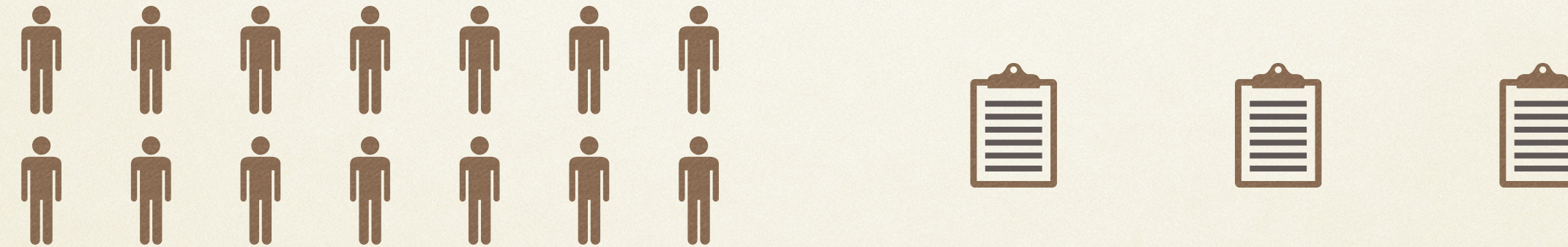
Security

GOALS OF ZKFOCIL



Security

ILs Unlinkable + Hidden Committee

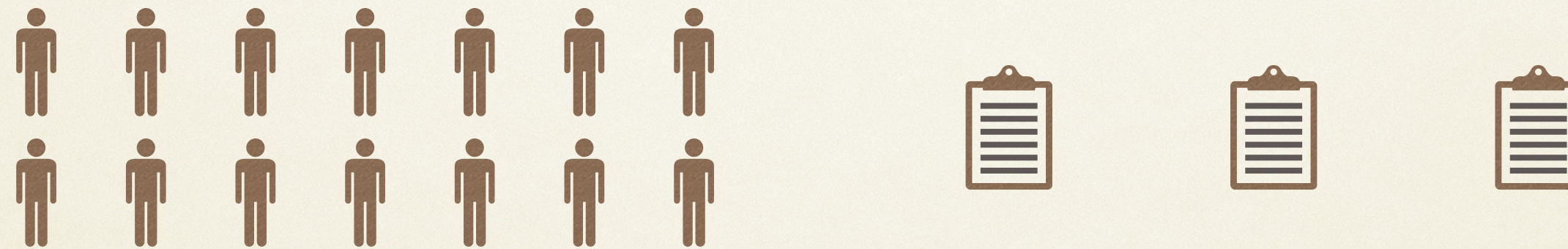


GOALS OF ZKFOCIL



Security

ILs Unlinkable + Hidden Committee



Unbiased Committee Election

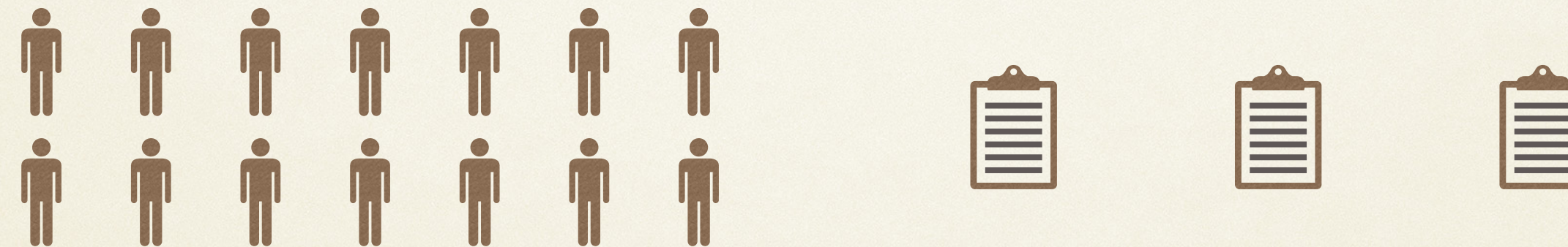


GOALS OF ZKFOCIL



Security

ILs Unlinkable + Hidden Committee



Unbiased Committee Election



Applicability

Efficiency

Conceptual Simplicity

Minimal Protocol Changes

TOWARDS A CONSTRUCTION



TOWARDS A CONSTRUCTION



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TOWARDS A CONSTRUCTION



Linkable Ring Signatures



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TOWARDS A CONSTRUCTION



Classical Constructions



Linkable Ring Signatures



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TOWARDS A CONSTRUCTION



Classical Constructions

Not efficient
For our application

Linkable Ring Signatures

zkFOCIL

TOWARDS A CONSTRUCTION



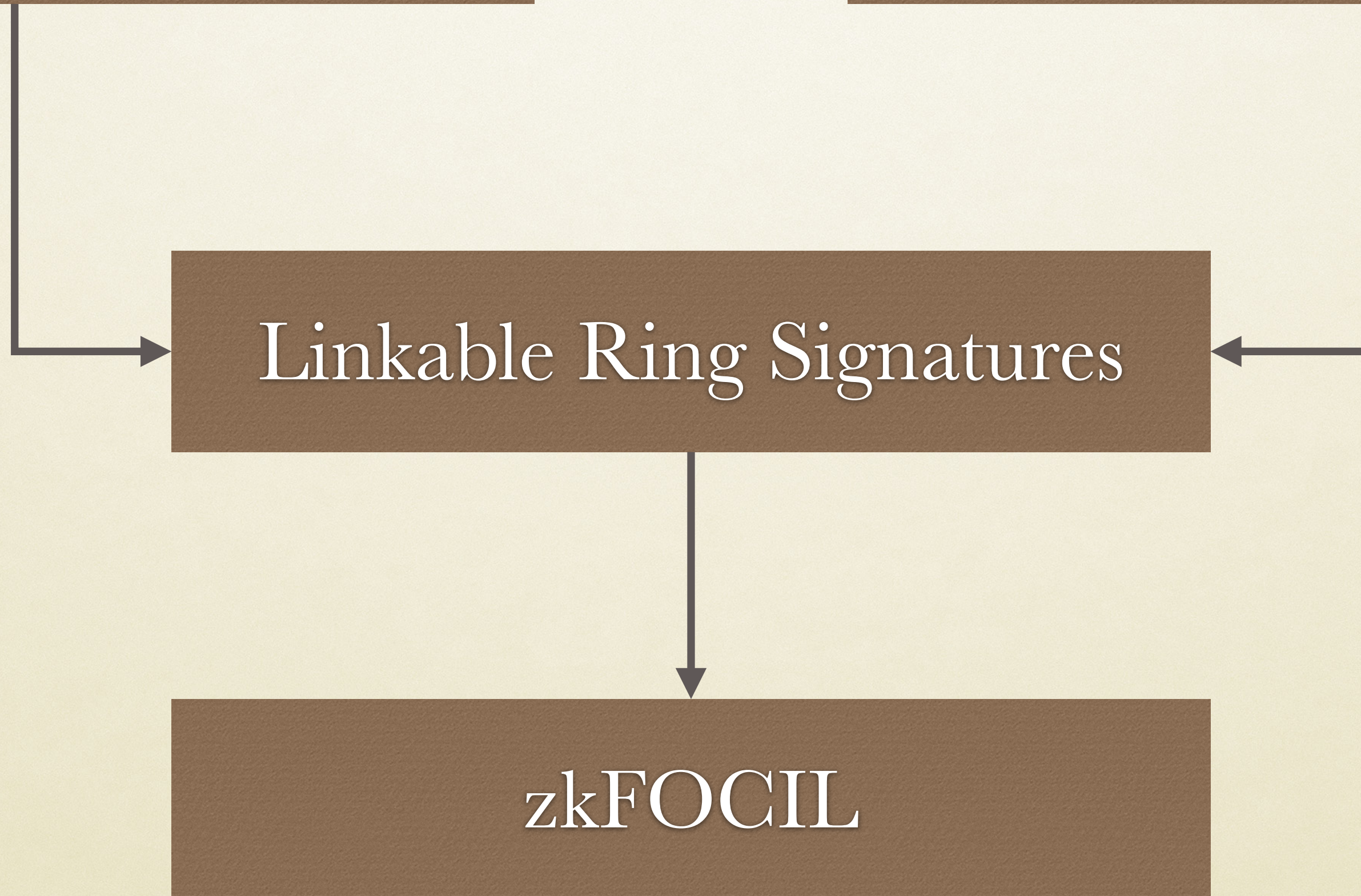
Classical Constructions

Generic zkSNARKs

Not efficient
For our application

Linkable Ring Signatures

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TOWARDS A CONSTRUCTION



Classical Constructions

Generic zkSNARKs

Not efficient
For our application

Almost Efficient
Enough

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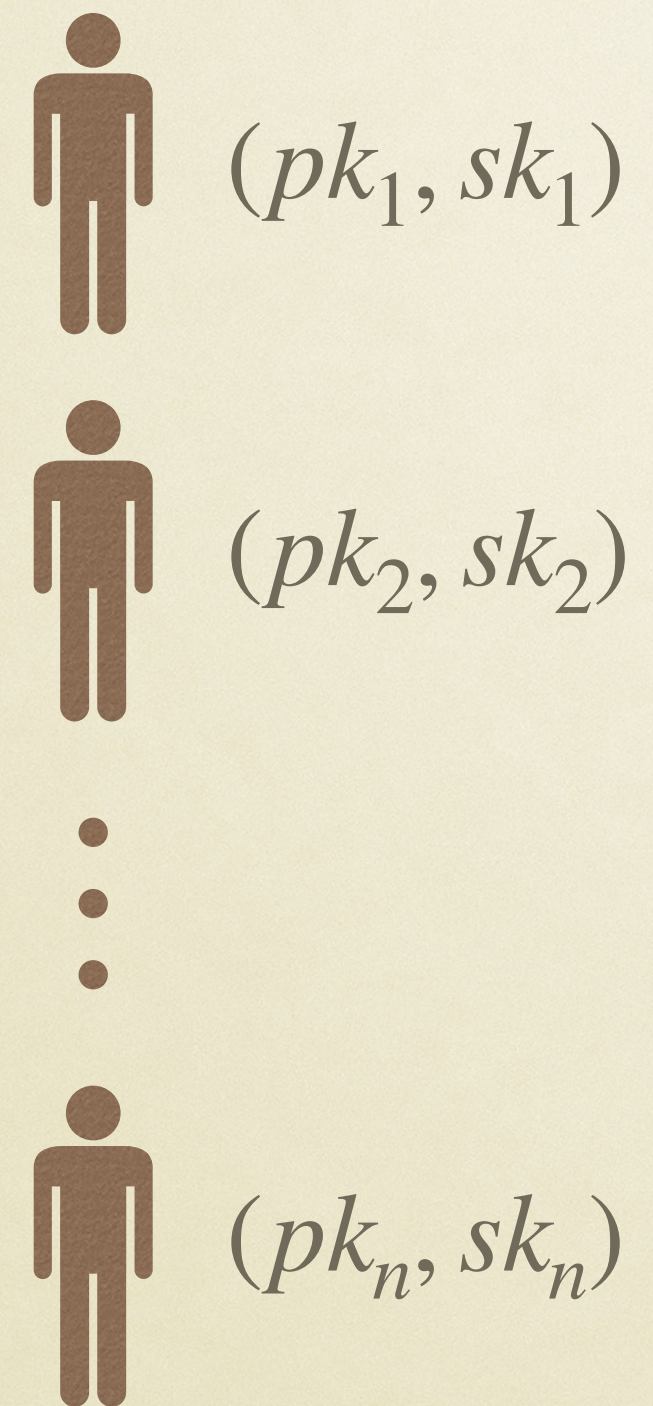
Linkable Ring Signatures

This talk

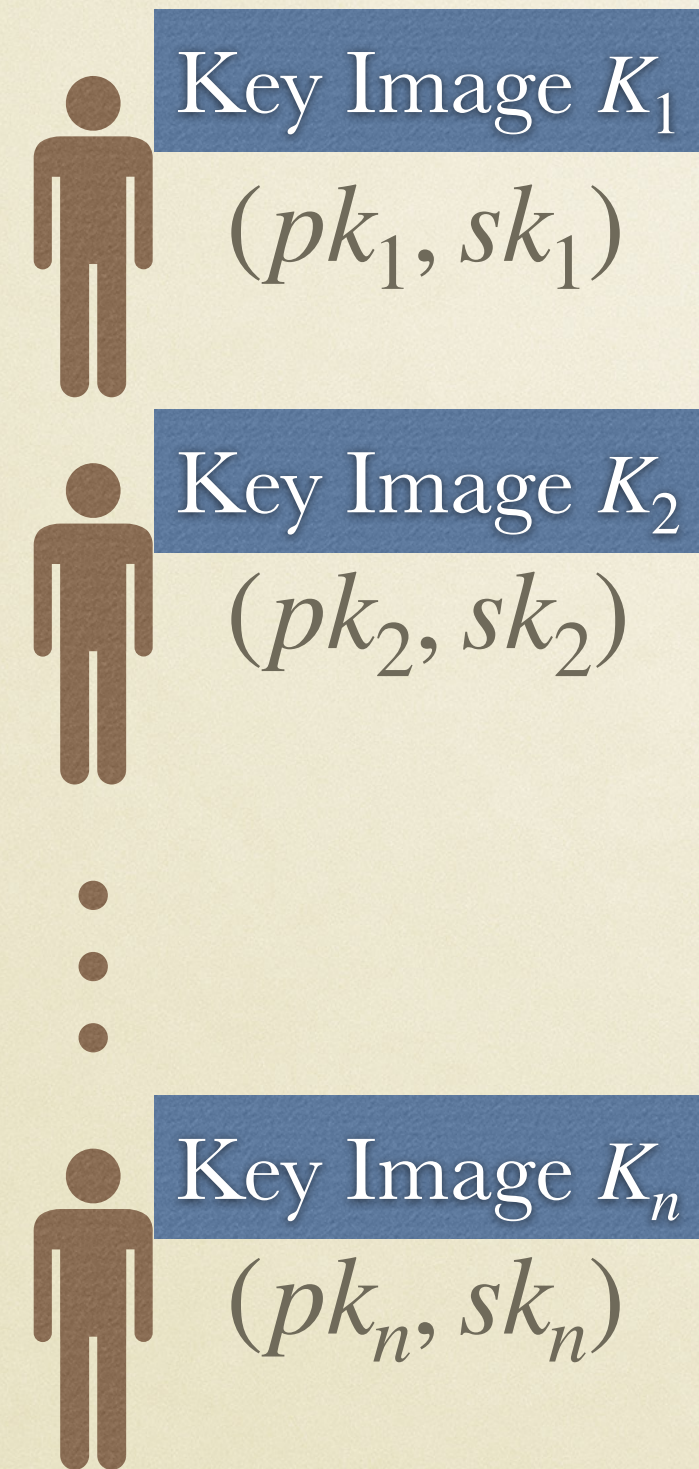
zkFOCIL



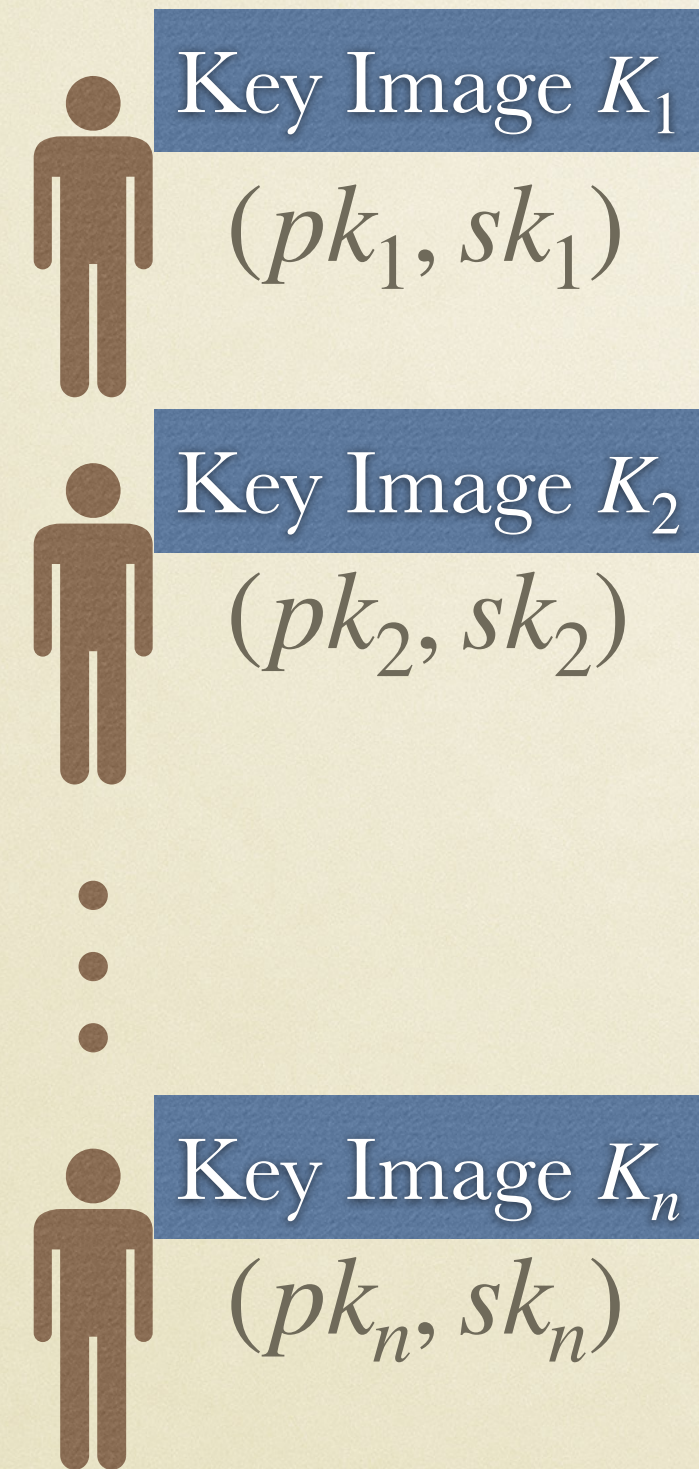
LINKABLE RING SIGNATURES



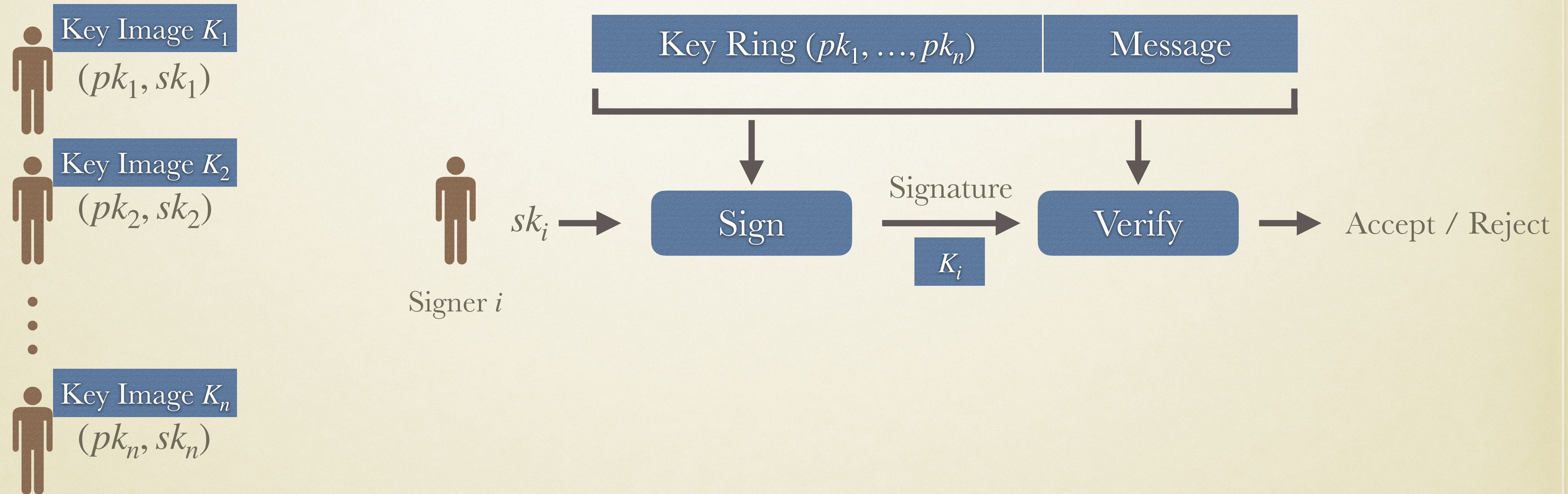
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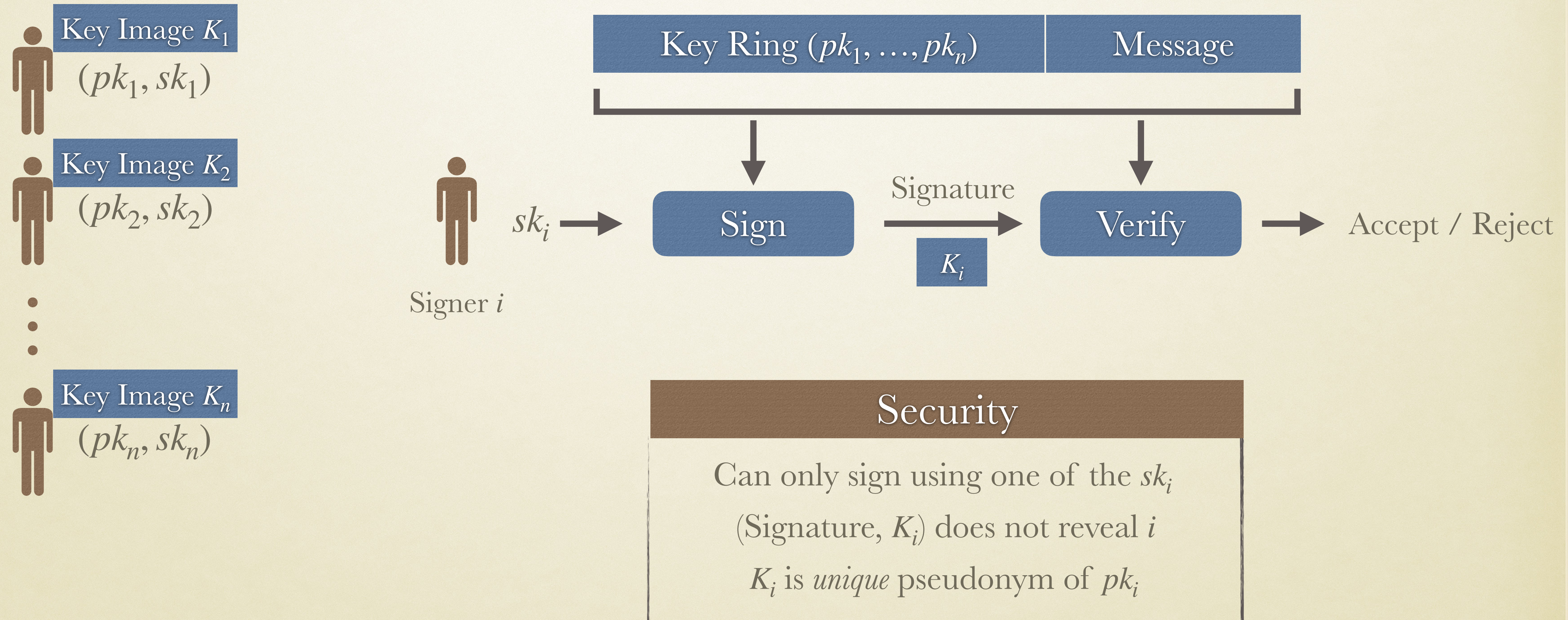
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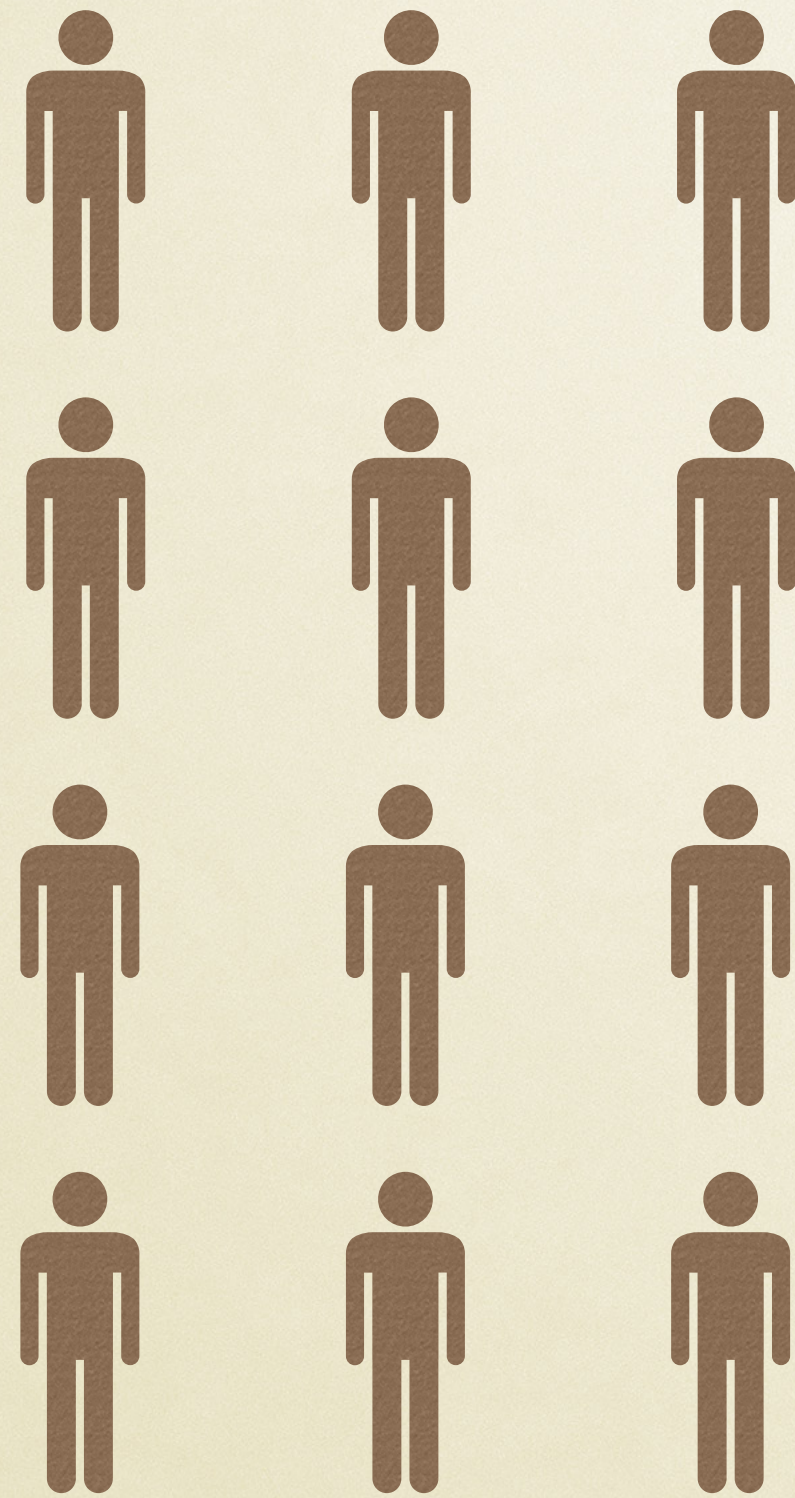
LINKABLE RING SIGNATURES



LINKABLE RING SIGNATURES IN FOCIL



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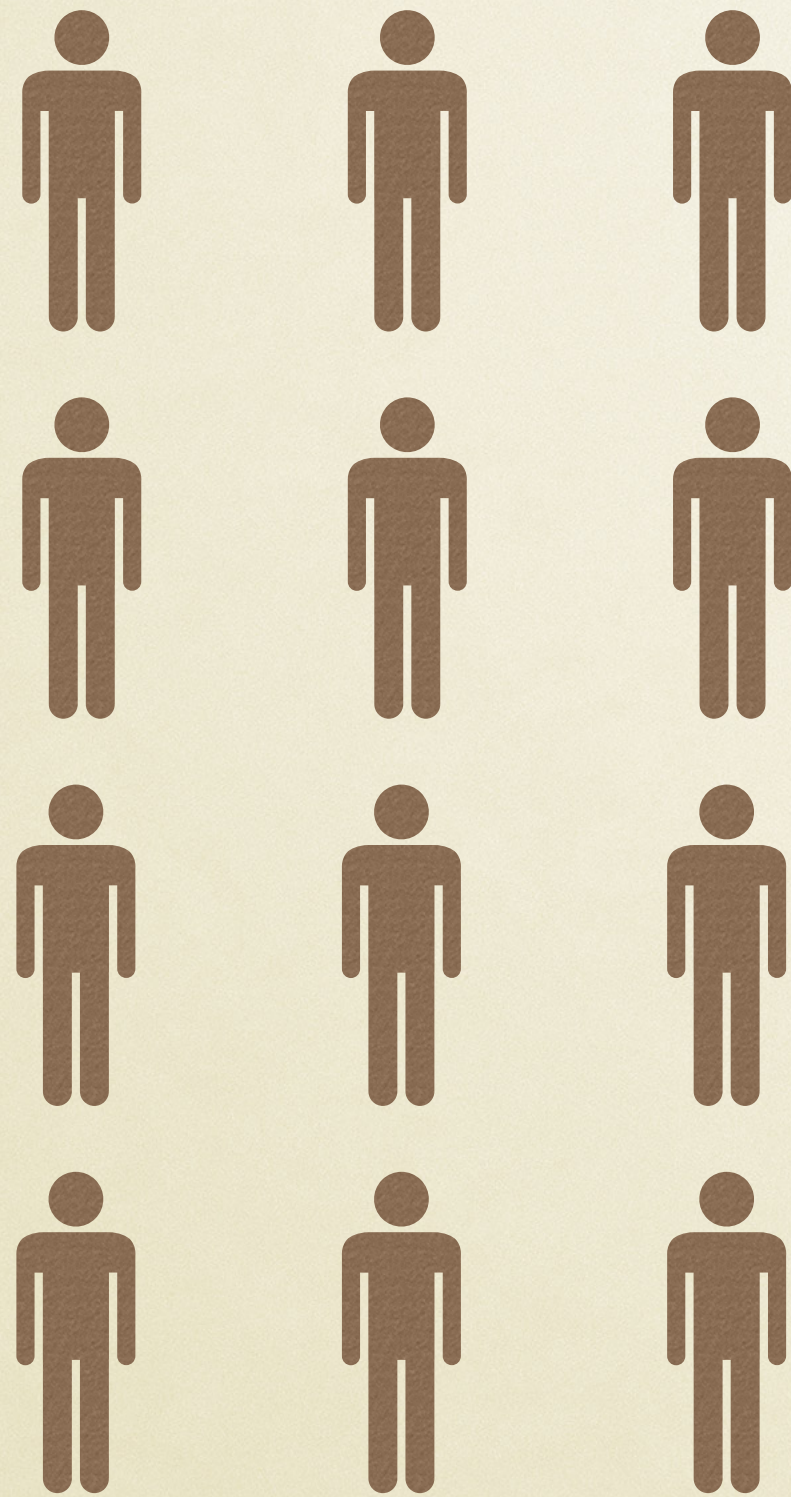


Validators

LINKABLE RING SIGNATURES IN FOCIL



Key Ring

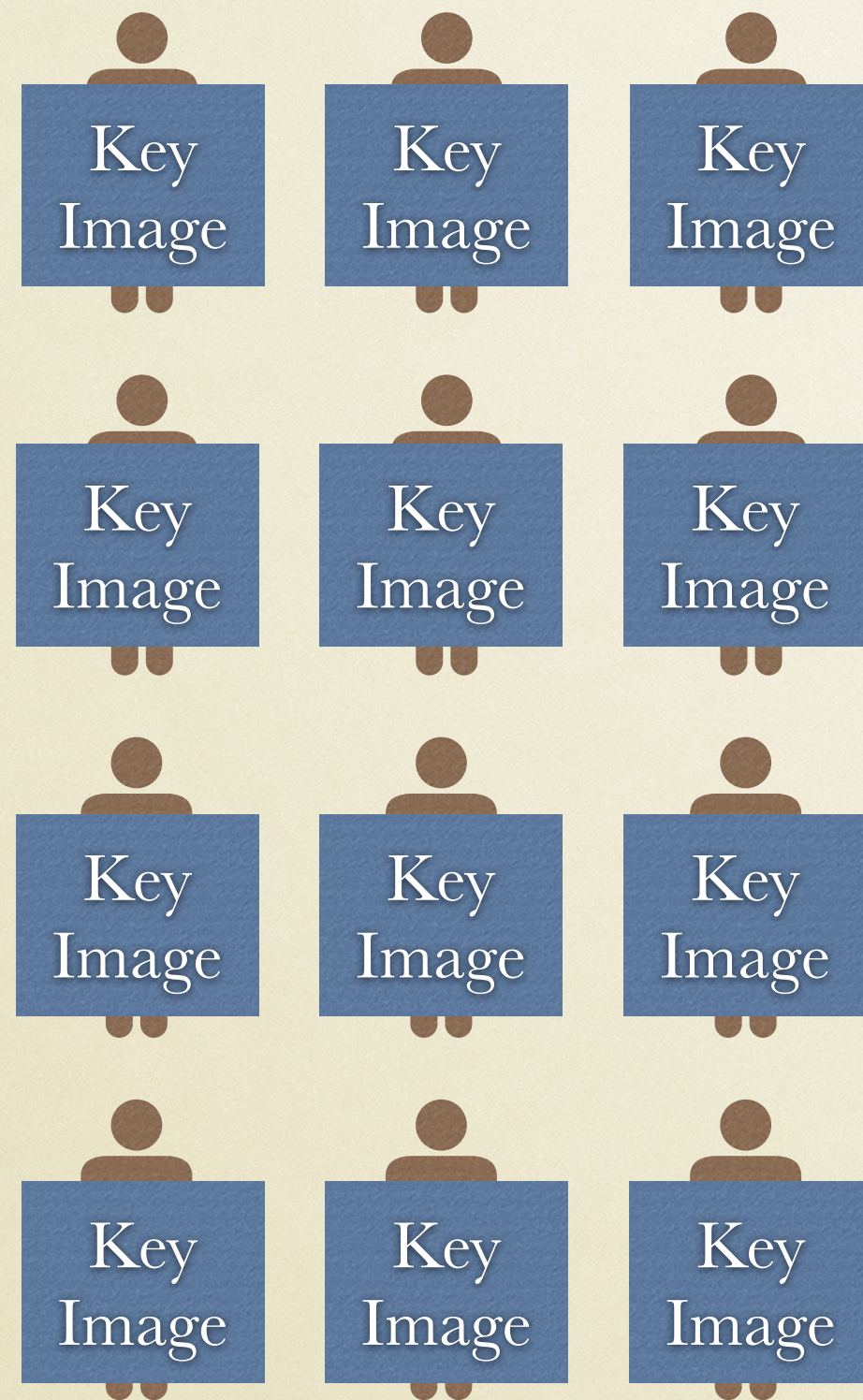


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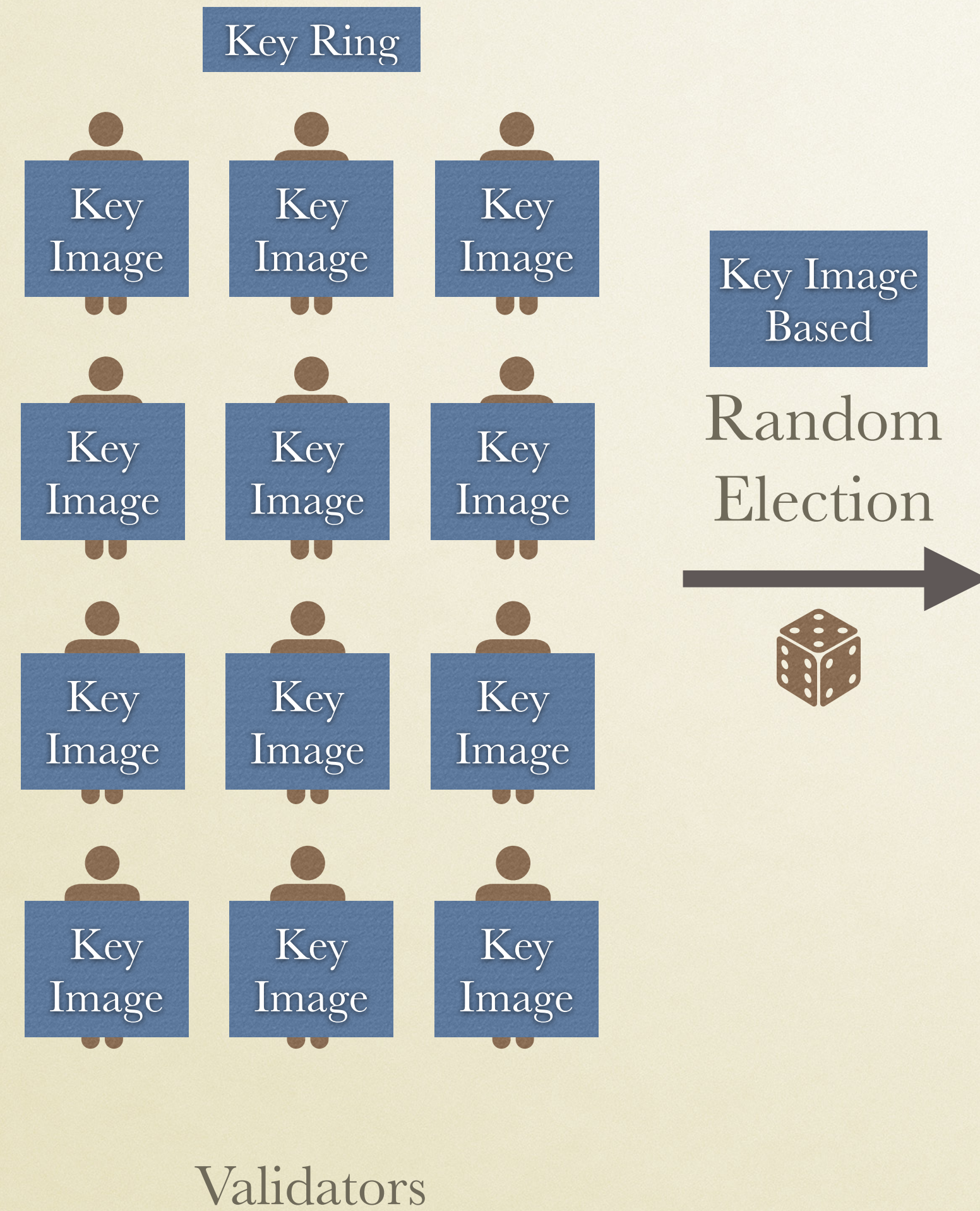


Key Ring

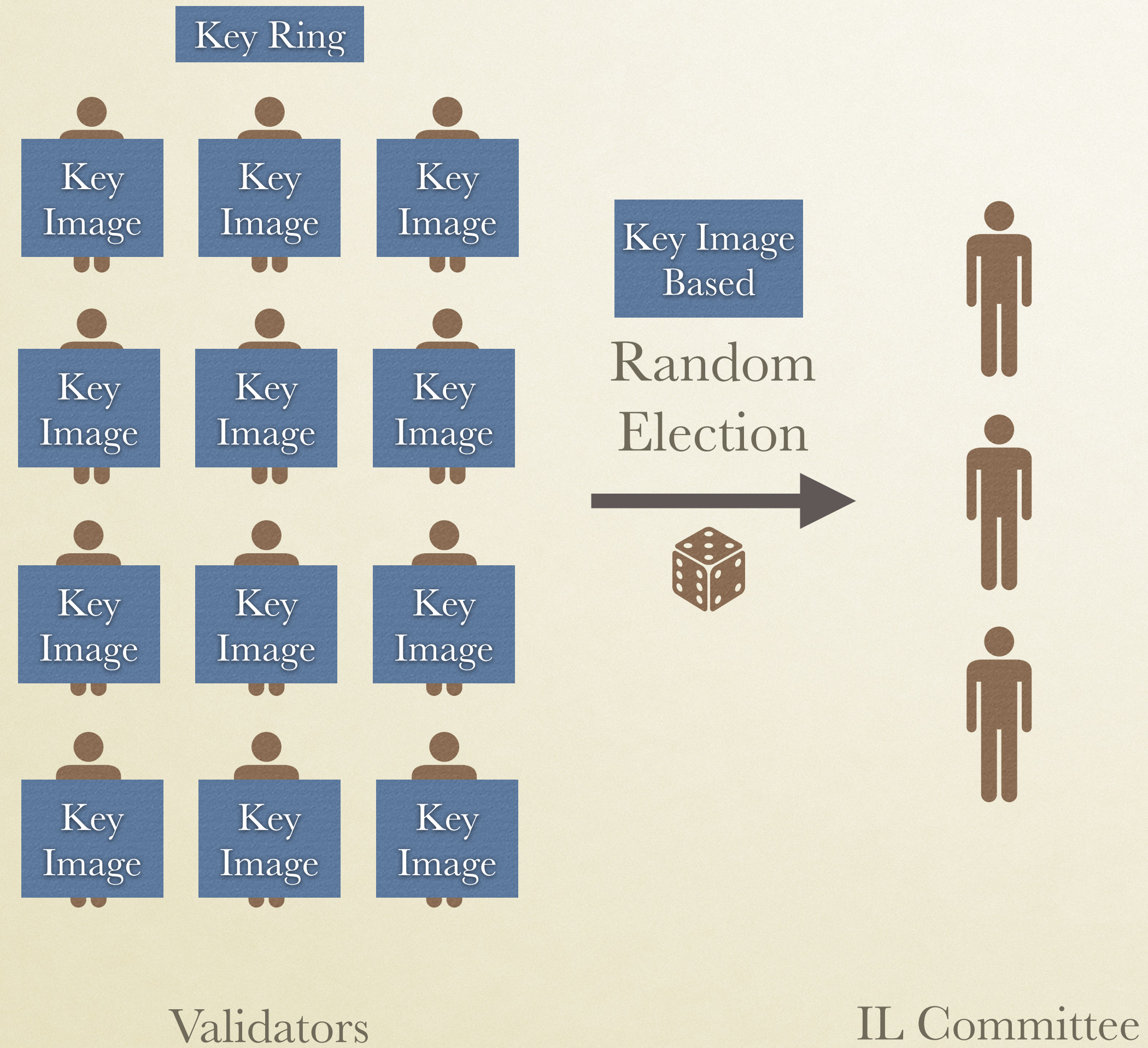


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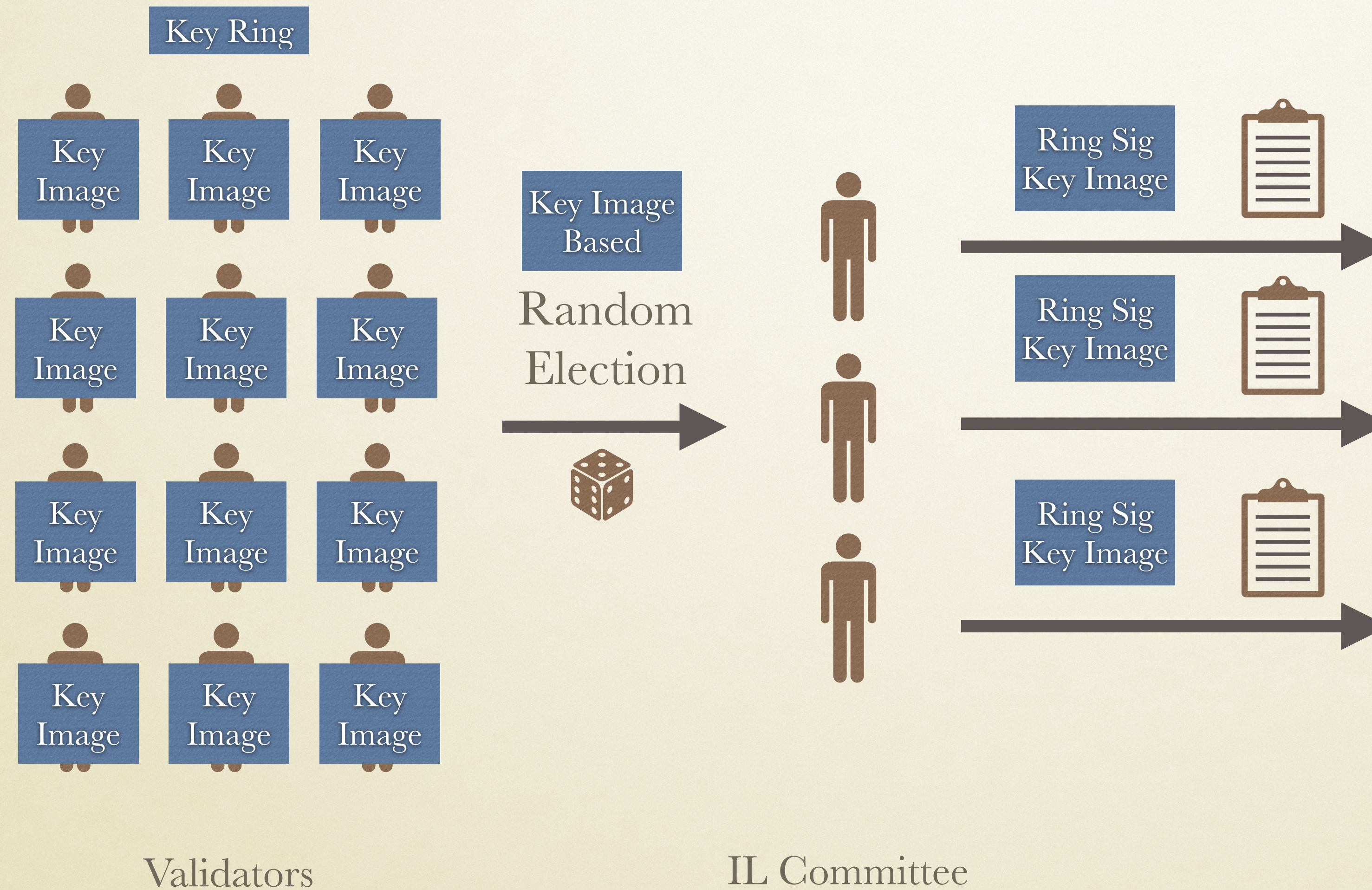
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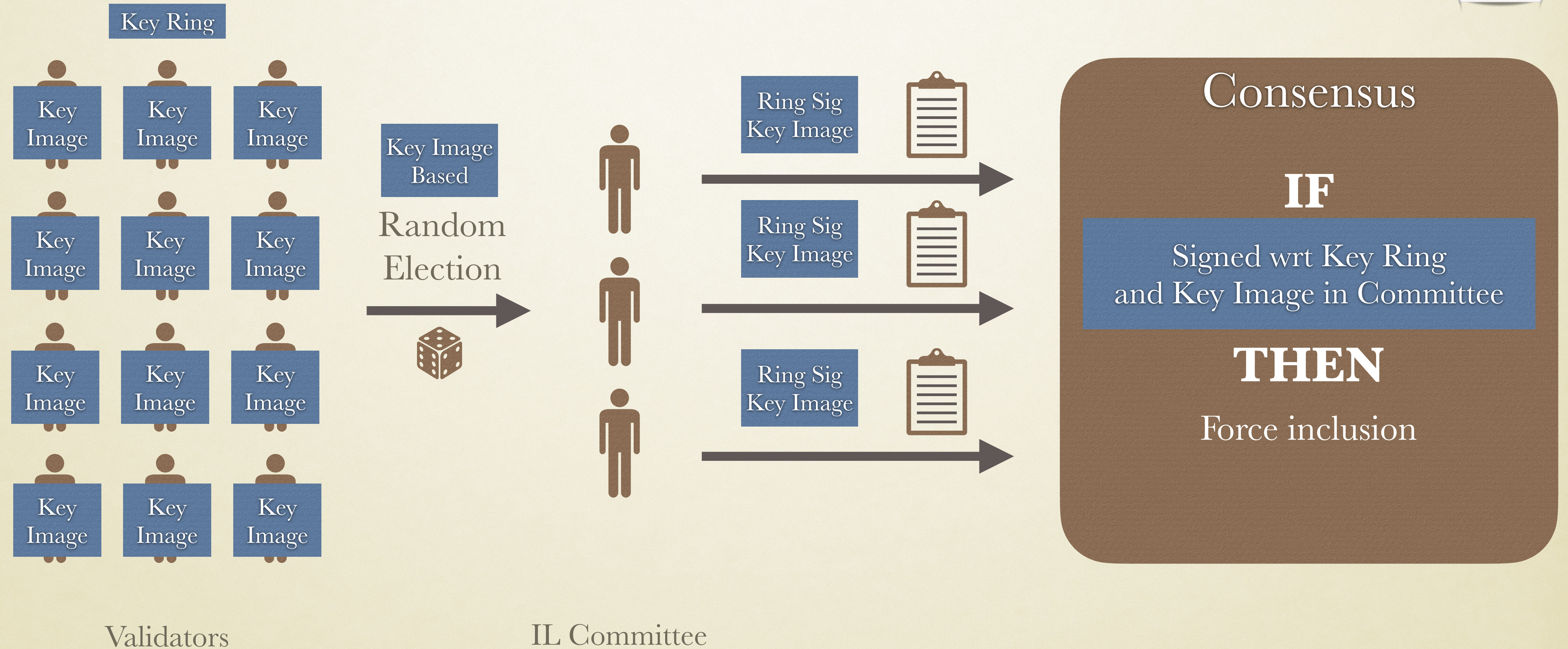
LINKABLE RING SIGNATURES IN FOCIL



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TOWARDS A CONSTRUCTION



Classical Constructions

Not efficient
For our application

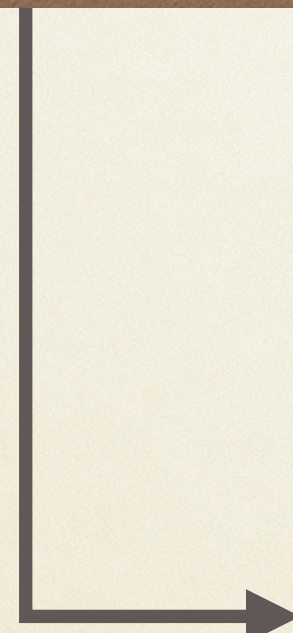
Generic zkSNARKs

Almost Efficient
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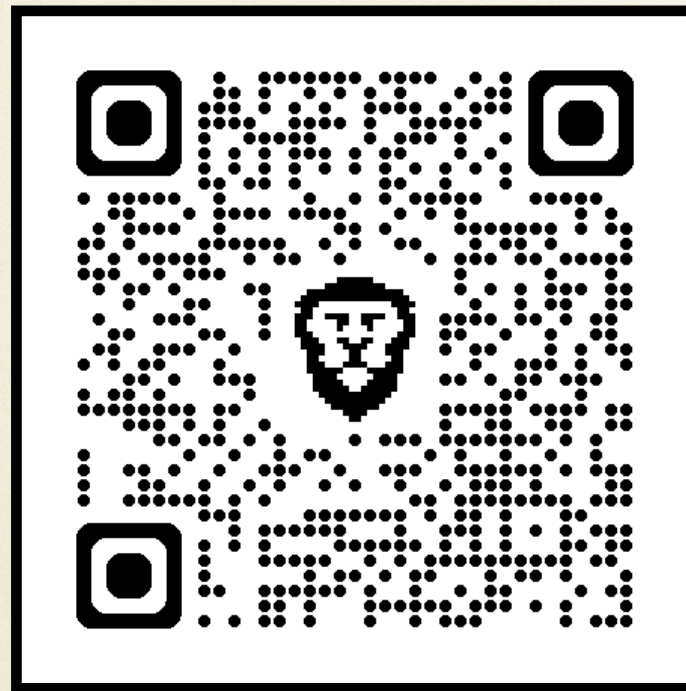
Linkable Ring Signatures

This talk

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EFFICIENCY (FROM GENERIC ZKSNARKS)



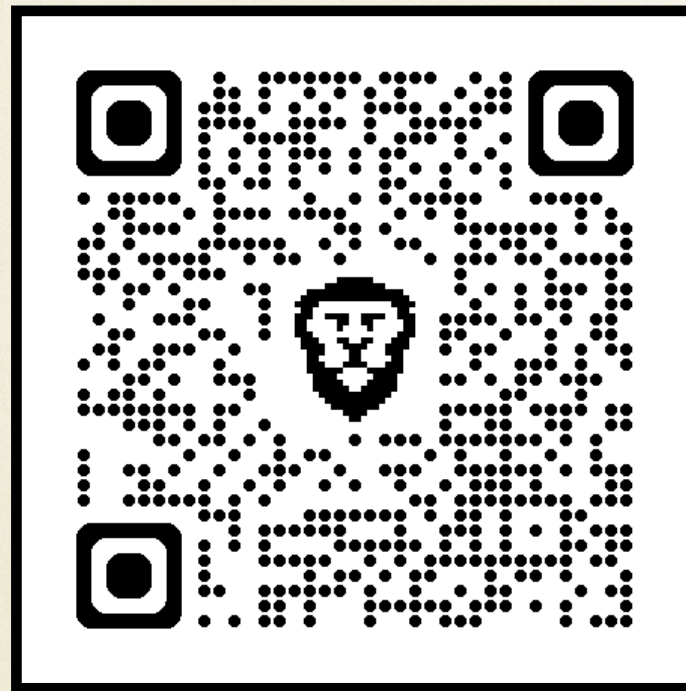
Prototype Implementation

by Shreyas Londhe & Suyash Bagad

* Sig + Key image size $\sim$ 500 Byte

* BN254+IPA / SECP256k1+IPA

EFFICIENCY (FROM GENERIC ZKSNARKS)



Prototype Implementation

by Shreyas Londhe & Suyash Bagad

Key Image + Proof

< 2.5 s

Sign IL

~ BLS sig

Verify Sig + Key Image

87 ms

* Sig + Key image size ~ 500 Byte

* BN254+IPA / SECP256k1+IPA

NEXT STEPS

NEXT STEPS

More efficient Construction

NEXT STEPS

More efficient Construction

zkFOCIL Combinatorics

NEXT STEPS

More efficient Construction

zkFOCIL Combinatorics

Integration into FOCIL

THANK YOU

